

A lattice law with a smooth Cramér rate

Disproof of conjecture 00000002802

ColdSmile 4 October 2026

Abstract

The fair Bernoulli probability law is a nondegenerate lattice law with every exponential moment. Its actual logarithmic moment generating function has a convex dual equal to $\log 2 + x \log x + (1 - x) \log(1 - x)$ for $0 < x < 1$, and this real representative is C^∞ there. Thus interior twice differentiability does not imply a non-lattice law. We distinguish ordinary support, whose interior is empty, from convex support and the effective domain, whose interiors are both $(0, 1)$. Lean verifies the necessary-implication failure for all three explicit readings.

1 Statement and logical target

The complete English statement in the repository is:

Definition: Cramér's theorem: deviation rates of independent sums (convex-dual form). Conjecture: Second-order differentiability of the Cramér functional: the rate function is twice differentiable in the interior of the support if and only if the underlying distribution is non-lattice, the quadratic coefficient being the variance functional; the lattice correction is a superposed periodic function. (Cramer second-order differentiability criterion)

Both the English and Chinese versions were read in full at the source revision cited below. Write D for the asserted twice differentiability and N for the non-lattice property. The equivalence $D \iff N$, or the Chinese grouping $D \iff (N \wedge V)$ with its additional variance assertion V , necessarily implies $D \implies N$. The example below refutes that implication. No interpretation of the unspecified variance functional or periodic correction is needed; neither those clauses nor the converse implication is separately settled here.

2 The probability law and its actual convex dual

On the Borel real line, let

$$\mu = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1.$$

This is a probability measure, with mass $1/2$ at each of 0 and 1 , and it is not almost surely constant. A law is *lattice* if, for some $a \in \mathbb{R}$ and $h > 0$, it is concentrated on $a + h\mathbb{Z}$. Our law is lattice with $a = 0$ and $h = 1$. Non-lattice means the negation of this condition.

For a probability law ν , put

$$T_\nu = \{t \in \mathbb{R} : y \mapsto e^{ty} \text{ is integrable against } \nu\}, \quad \Lambda_\nu(t) = \log \int_{\mathbb{R}} e^{ty} d\nu(y),$$

where Λ_ν is used only for $t \in T_\nu$. Define the extended-real convex dual by

$$I_\nu(x) = \sup_{t \in T_\nu} \{tx - \Lambda_\nu(t)\}. \quad (1)$$

The restriction avoids the totalized real integral at nonintegrable parameters. The zero tilt is integrable for any probability law and gives $I_\nu(x) \geq 0$. For μ , all real tilts are integrable, so the supremum is over $\mathbb{R}$. Direct integration gives

$$M_\mu(t) = \int e^{ty} d\mu(y) = \frac{1 + e^t}{2}, \quad \Lambda_\mu(t) = \log \frac{1 + e^t}{2}. \quad (2)$$

3 Computing the supremum, rather than assigning a formula

Fix $0 < x < 1$ and define the auxiliary real expression

$$F(x) = \log 2 + x \log x + (1 - x) \log(1 - x).$$

We prove that the actual supremum (1) equals $F(x)$.

Upper bound for every tilt. Apply concavity of $\log$ with weights $x, 1 - x$ and positive arguments e^t/x and $1/(1 - x)$. Their weighted sum is $e^t + 1$, so

$$\begin{aligned} x \log \frac{e^t}{x} + (1 - x) \log \frac{1}{1 - x} &\leq \log(e^t + 1), \\ xt - x \log x - (1 - x) \log(1 - x) &\leq \log(e^t + 1). \end{aligned}$$

Since $\log((1 + e^t)/2) = \log(1 + e^t) - \log 2$, rearrangement yields

$$tx - \Lambda_\mu(t) \leq F(x) \quad (t \in \mathbb{R}). \quad (3)$$

A tilt attaining equality. Set $t_x = \log x - \log(1 - x)$. Then

$$e^{t_x} = \frac{x}{1 - x}, \quad \frac{1 + e^{t_x}}{2} = \frac{1}{2(1 - x)}.$$

Substitution into the dual objective gives

$$t_x x - \Lambda_\mu(t_x) = x \log x - x \log(1 - x) + \log 2 + \log(1 - x) = F(x).$$

Together with (3), this proves

$$\boxed{I_\mu(x) = F(x) = x \log(2x) + (1 - x) \log(2(1 - x)), \quad 0 < x < 1.} \quad (4)$$

The equality in Lean is an equality in `EReal` between the actual supremum and the coercion of this real expression.

4 The effective-domain interior

Let $E_\mu = \{x : I_\mu(x) < +\infty\}$. Because $I_\mu \geq 0$, this set really is its finite effective domain. Equation (4) proves $(0, 1) \subseteq E_\mu$.

For $t \leq 0$, one has $e^t \leq 1$, so $\Lambda_\mu(t) \leq 0$. If $x < 0$, then for any $A \in \mathbb{R}$ the tilt $t = (|A| + 1)/x < 0$ satisfies

$$tx - \Lambda_\mu(t) \geq |A| + 1 > A.$$

Thus $I_\mu(x) = +\infty$. For $t \geq 0$, one has $1 \leq e^t$, hence $(1 + e^t)/2 \leq e^t$ and $\Lambda_\mu(t) \leq t$. If $x > 1$, choose $t = (|A| + 1)/(x - 1) > 0$; then

$$tx - \Lambda_\mu(t) \geq t(x - 1) = |A| + 1 > A,$$

again giving $I_\mu(x) = +\infty$. Consequently

$$(0, 1) \subseteq E_\mu \subseteq [0, 1], \quad \text{int } E_\mu = (0, 1).$$

Only this sandwich and the interior equality are needed. Endpoint values and the equality of the entire effective domain with $[0, 1]$ are not claimed as results of this formal development.

5 Smoothness and the three support readings

On $(0, 1)$ both x and $1 - x$ are positive. The logarithm is smooth there, so sums, products and composition show that F is C^∞ on $(0, 1)$. By (4), the actual rate has this finite smooth real representative on that open interval. This avoids using the totalized real projection of an infinite extended-real value.

The ordinary topological support is defined by the usual open-neighborhood condition: $x \in \text{supp } \mu$ if every open neighborhood of x has nonzero measure. Both atoms are in the support because their singleton masses are positive. Its complement has measure zero, and is open, so

$$\text{supp } \mu = \{0, 1\}, \quad \text{int supp } \mu = \emptyset.$$

Its convex hull is $[0, 1]$. In particular, these three readings are explicitly different:

Region used for differentiability	Interior	Certificate
Ordinary topological support	$\emptyset$	Vacuous
Convex hull of that support	$(0, 1)$	Smooth finite representative
Effective domain of the actual rate	$(0, 1)$	Smooth finite representative

The last two interiors are nonempty, for example at $1/2$. Ordinary support is never silently replaced by convex support.

The exact differentiability predicate. For an extended-real function I on an open set S , the formal predicate `RateTwiceDifferentiableOn`, applied to I and S , asserts the existence of a real function f such that f and its ordinary derivative f' are differentiable on S , and $I(x) = (f(x) : \text{EReal})$ for all $x \in S$. Since S is open, equality holds on a neighborhood of each point and identifies the local differentiability of the rate itself.

The stronger predicate `RateC2On` requires a C^2 real representative. Lean proves its sufficient implication to the twice-differentiability predicate on open sets. Mere twice differentiability is not identified with C^2 regularity. The proof of smoothness uses `ContDiffOn` with order ∞ (all finite orders); a C^2 certificate then follows. The empty-set certificate is recorded separately and vacuously.

Proposition 1. *The source’s necessary implication from interior twice differentiability to a non-lattice law fails for each of the ordinary-support, convex-support and effective-domain interiors.*

Proof. In every reading, μ satisfies the corresponding differentiability assertion, as proved above, but is a lattice law. It is also a nondegenerate probability law with every exponential moment. Thus the necessary implication fails even on this restricted class of particularly regular probability laws. Those restrictions strengthen the counterexample; they are not additional assumptions attributed to the source. $\square$

Formal scope. The development constructs the actual probability-law convex dual specified in the source and refutes its analytic differentiability criterion. It does not formalize an infinite independent process, the full Cramér large-deviation theorem, or probability asymptotics. The definitions and proof do not rely on any of those unformalized results. The standard Bernoulli convex-dual formula and the conventional interior-smoothness context can also be found in Fischer’s lecture notes, §2.3 [2]; the identities used here are proved directly.

6 Lean correspondence and reproducibility

All named declarations below belong to the namespace `Conjecture2802`. The Lean project is in `lean/`; each module is included in the default library build.

Law.lean Defines `fairBernoulli`, `LatticeLaw` and `topologicalSupport`; proves the probability instance, both masses, nondegeneracy, exponential integrability, actual moment/cumulant formulas, and the ordinary/convex support identities.

Formula.lean Proves `dual_objective_le_binaryRate` and `dual_objective_optimalTilt`: the upper bound and the attaining tilt.

Transform.lean Defines `cramerRate` and `effectiveDomain`; proves `cramerRate_nonneg`, `cramerRate_fairBernoulli`, exterior infinity, the domain sandwich and `interior_effectiveDomain_fairBernoulli`.

Regularity.lean Defines both regularity predicates; proves `binaryRate_contDiffOn` and the open-set bridge `RateC2On.twiceDifferentiableOn`.

Conjecture2802.lean Supplies the explicit counterexample and the three precise necessary-implication refutations. The combined conclusion is `necessary_criterion_fails_all_three_readings`; the bundled witness is `explicit_counterexample`.

Check.lean Prints the definitions, theorem types and axiom dependencies for the complete submitted declaration inventory.

Pinned environment. The project uses Lean 4.19.0 and Mathlib v4.19.0, with Mathlib commit `c44e0c8ee63ca166450922a373c7409c5d26b00b`. The included toolchain and lockfile fix the compiler and dependency revisions.

From the submission directory, the reproduction commands are:

```
cd lean
lake exe cache get
lake build
for f in Conjecture2802/Law.lean Conjecture2802/Formula.lean \
  Conjecture2802/Transform.lean Conjecture2802/Regularity.lean \
  Conjecture2802.lean Check.lean; do
  lake env lean -DwarningAsError=true "$f" || exit 1
done
```

Validation status. An independent fresh project build completed successfully on 4 October 2026 at 21:42:42 UTC. All six Lean sources were replayed with warnings treated as errors. The inventory contains 62 declarations: 13 definitions, 48 theorems and one named probability instance. All 49 theorem/instance type and transitive-axiom checks passed, with dependencies limited to the standard axioms `propext`, `Classical.choice` and `Quot.sound`. No admitted proofs, `native_decide`, custom axioms or other proof bypasses were found. The nine source/configuration hashes were unchanged; all nine pinned dependency checkouts were tracked-clean and the compiler identity was checked before and after execution. The full records and file identities are supplied in `verification/` and `VERIFICATION.md`. No auxiliary numerical computation is needed: all mathematical identities and inequalities are established in Lean. Local verification is distinct from maintainer acceptance.

References

- [1] The Last Math Competition, conjecture 00000002802, English and Chinese versions, source revision `fe1d06d431b0591b65d759c60035b1d2e293a819`. [Repository source](#).
- [2] Markus Fischer, *Large deviations, weak convergence, and relative entropy*, revised 20 June 2013, §2.3, pp. 6–7. <https://www.math.unipd.it/~fischer/Didattica/WeakConvergenceRE.pdf>.